NIST HFBS

VME DAQ

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1 INTRODUCTION

The NIST HFBS VME data acquisition system is designed to acquire and histogram the counting rates of 22 neutron detectors (20 detector and 2 monitor channels) in dependence of a chopper wheel position, encoded via a grey code counter) and doppler drive data coming from 2 analogue outputs of a vibrometer. The front end components are a STR7201 multi scaler as counter, a STR7137 latch as trigger generator and latch and two OR VADM as ADCs. The readout of the front end components is performed by a STR8080 DSP as VME master and online histogrammer. Slow control of the data acquisition system is handled by a Heurikon VME CPU running under OS9 and with acces to a 3 GB SCSI disk drive in VME housing. The OS9 system is also in charge of the setup of the front end modules during the run setup phase. Run control and display of the histograms are performed on a LINUX host workstation (PC), data and run control information exchange taking place via an Ethernet socket communication and OS9 memory modules. The data transfer between the VME CPU and the DSP is established via a VSB connection to decouple the fast VME readout of the frond end modules from the transfer of histogram data. During data acquisition the VME CPU does not access the VME bus to avoid delays introduced by arbitration for the bus.

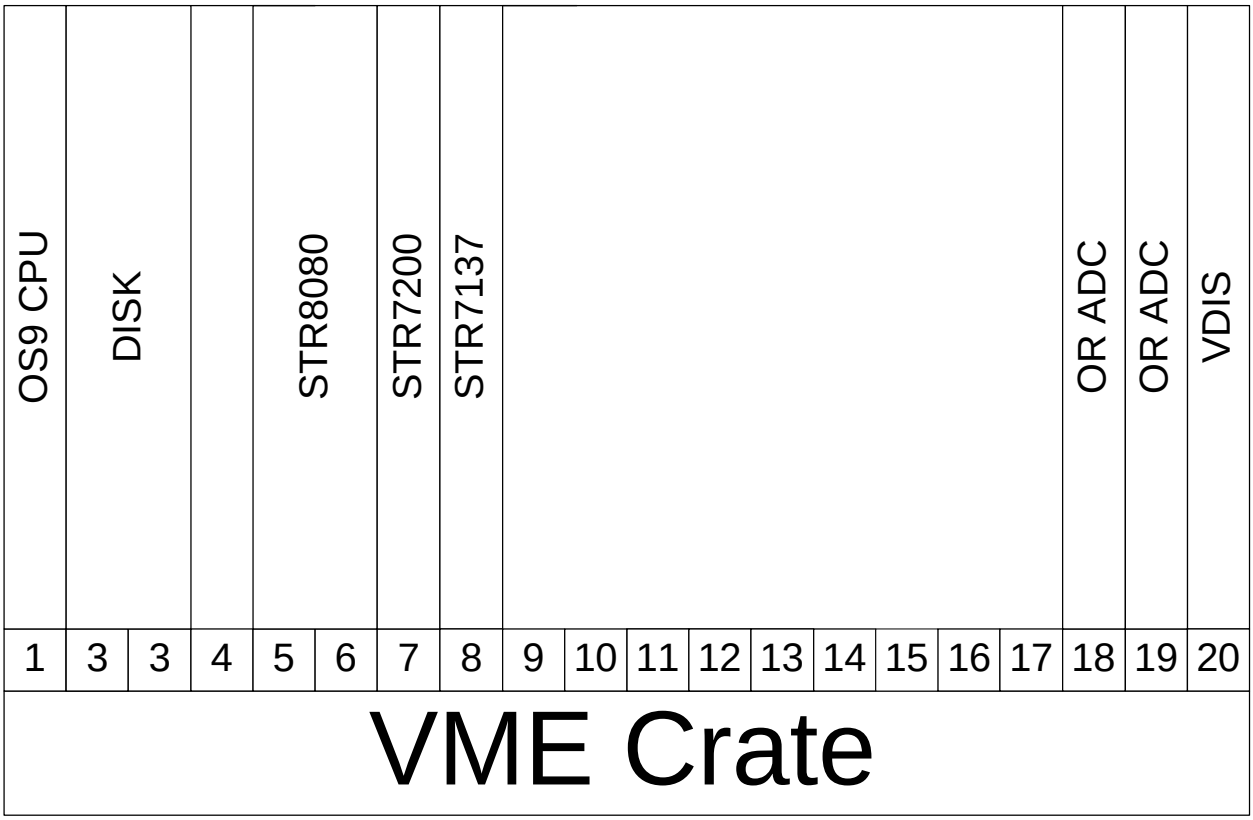
A three pointer (one write and two readout) readout pipeline between histogram pointer generation and actual histogram filling is established to take neutron time of flights for the counters and reference counters into account. Background compensation can be done by evaluation of the beam on/off histograms filled on the base of lookup tables for the chopper wheel position.

2 STANDARD OPERATION

The VME CPU is set to auto boot mode, hence it boots up as soon as the VME crate power is turned on. The boot procedure can be verified by the read HDD LED being active during the startup procedure. As soon as the VME system is up you can log on via telnet HVME from rrdswell or via a VT100 terminal connecting to one of the VME CPUs two serial ports. Login as root (user: super, pw: user) and type **chd nist/daq.** By typing **hfbsdaq** data acquisition on the VME side is started and waits for a socket connection from a host system sending commands. The startup of hfbsdaq can be included into the boot procedure as the system is moving from development to routine data acquisition mode.

3 VME SETUP

The drawing below illustrates the VME setup.



3.1 BASE ADDRESS SETTINGS

The following table shows the base addresses of the VME components and the VSB side of the 8080 DSP.

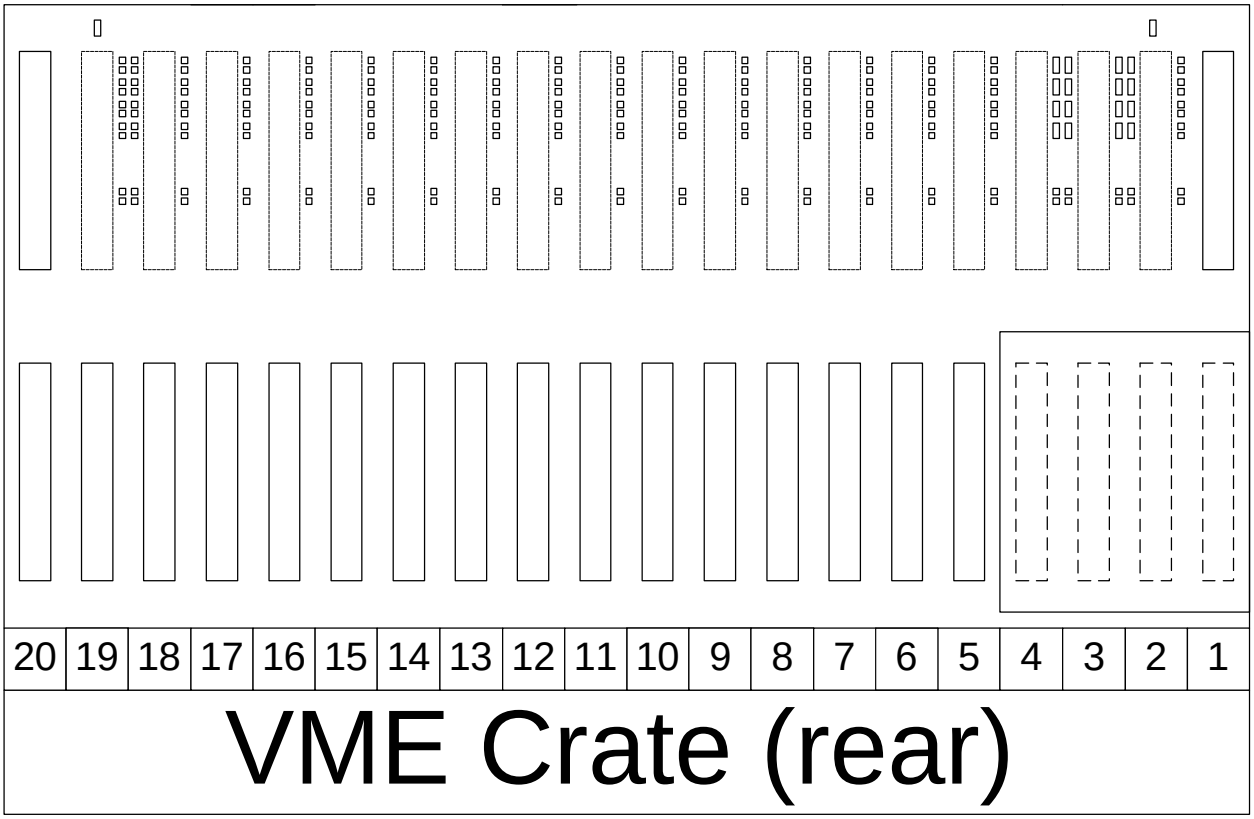
Module	Address
VDIS	0x40001000
VADM 1	0xE000
VADM 2	0xF000
STR7201	0x43508000
STR7137	0x43509000
STR8080	0x0F000000

3.2 VSB BACKPLANE AND BUS

A four slot backplane is installed onto the rear of the P2 backplane. It busses slots one through four of the P2 with the VSB. It is needed to transfer the histogrammed data from the DSP to the VME CPU and to ensure proper termination of the VSB (ie. It was found, that the system does not work properly with a cable backplane), as the Heurikon and the 8080 do not provide proper termination of the bus. The VSB bus is routed from the Heurikon in slot one of the crate through the backplane to slot four and from the front of the P2 connector to the STR8080 front panel VSB connector via a ribbon cable.

3.3 VME BUS GRANT (BG)

As the Heurikon is located in slot one of the VME crate and the STR8080 is in slot 5 and the VME slots in between are not used by VME modules (the SCSI disk is taking its power only from VME), the Bus Grant jumpers have to be set on the rear of the backplane to allow the STR8080 to get VME bus mastership. As no VME Interrupts are used in the current setup, no jumpers are installed on the IRQ daisy chain pins.



4 3 SYSTEM CABLING

4.1 VME CPU (HK4F)

The CPU is connected to the VME SCSI disk by a ribbon connector with one 50 pin female CENTRONICS connector on the disk side and a high density SCSI connector on the CPU side. Via a second ribbon cable with a female 15-pin DSUB connector on the transceiver side and a high density connector on the CPU side the Ethernet port is connected to the AUI side of an Ethernet transceiver.
If the boot setup is to be changed, a terminal can be connected to the CPU via a ribbon cable with 2 female 25-pin DSUB connectors on the terminal side and one high density serial connector on the CPU side. After disconnecting the SCSI cable from the Disk or CPU the boot menu will be invoked upon power up.

4.2 STR8080

The STR8080 is connected to the VSB bus via a ribbon cable. 64 pins (row A and C) of a 96 pin female connector plugging into the front panel of the DSP are connected to the P2 backplane after a female/male transition (the transition is needed to work around a swapping of rows A and C).

4.3 TRIGGER/TIMING

Four of the control signals of the STR7137 are used as ECL outputs with the selected LCA design. The are connected to an internal clock, which can be set in 100 ns steps. Three of these are used to distribute the readout clock to strobe the latch itself, to give the next start to the STR7201 and to generate the readout interrupt to the STR8080 DSP. The connections are done by three short (1 pair) cables. Care has to be taken to maintain the proper polarity.

4.4 INPUT SIGNALS

4.4.1 Scaler

The ECL counter signals are connected to the two 34 pin headers of the STR7201 multiscaler via two ribbon cables, the lower connector being used for channels 1-16. Refer to the STR7201 manual for the input polarity.

4.4.2 Latch

The ECL latch bits are connected to the lower 34 pin header of the STR7137 latch via a ribbon cable. Refer to the STR7137 manual for the input polarity. To be on the safe side bits except bits 0-11 are anded out by the DSP software (ie. floating unused bits will not cause a problem, but this has to be kept in mind for future possible upgrades).

4.4.3 ADCs

Refer to the OR computers VADM10 documentation for the input connector layout and input configuration. The ADC is set to a range of -10 ... +10 V. Channel zero of both ADCs is the only used channel in the present setup to allow for 100 KHz conversion in fast mode.

5 VME DISK

The OS-9 system disk is a Quantum Fireball 3.2 GB SCSI drive mounted in a 2 slot wide VME enclosure. +5 V+12 V and ground are the only lines connected to the bus. The presence of power is signalled by a green (+12 V) and a yellow (+5 V) LED. A red LED indicates, that the harddisk is accessed by the VME CPU. The Disk is connected to the SCSI bus via a female 50 Pin Centronics connector.

6 SOFTWARE

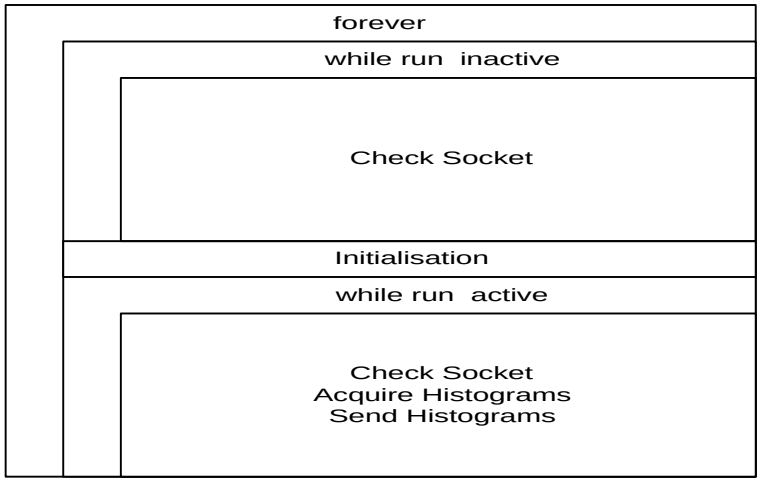
6.1 OS9 SIDE

6.1.1 Directory Structure on the OS9 System

The VME SCSI disk is mounted under /h0. All files for the HFBS are grouped under /h0/NIST.

6.1.2 Main Program Flowchart

The mainprogram has the following very simple structure:



6.1.3 Lookup tables

6.1.3.1 PST

The program /h0/nist/daq/genlook.c is a simple example program for the generation of a lookup table /h0/nist/dsp/lookup.tbl. Beam on positions have to be set to the base address of the beam on histos, beam off positions to the address for the beam off histos (ie. beam on +0x5000). The first 4096 entries are for the detectors, the next 4096 entries for the monitor counters. The lookup tables are loaded to the DSP at program start independent of the acquisition mode.

6.1.3.2 Energy and other

The energy (histogram bin) is computed according to the formula E=a*v+b*v*v. V is obtained from the ADC of the vibrometer. Although looking pretty simple, this computation including ADC scaling and sign checking would take some 15 lines of DSP code. As the ADC is 12 bit (11+sign) only and the histogram bin supposed to fall in the range 0-1023 (ie. 10 bit) ist straightforward to use a lookup table to map the ADC reading to the histo bin. The program genadcl.c is designed to generate this table and to store it in /h0/nist/dsp/adclook.tbl, from where it is downloaded to the DSP at initialisation time.
Two lookup tables for Dms+ddop and Vbragg+v are generated by this program as well, they are stored under same location as top.tbl and botlook.tbl.

6.2 DSP SIDE

The DSP code is written in assembly language and assembled and linked with the Motorola CLASA development package using the batch file asm.bat in a dos shell. The syntax is ***asm <file>***, where <file> is the name of file.asm without extension (***asm nist*** assembles the file nist.asm eg.). The batch file also produces the downloadable (ASCII) ***<file>.dmp*** (***nist.dmp*** in the example) with the cofdmp utility.
The documentation for the 96002 DSP and its assembly language can be found on the documentation CDROM in PDF format.

6.3 LINUX SIDE

The files are stored in the hfbs home directory under the subdirecroties dsp, os-9 and source. The directory dsp hold the assembly source files teach.asm and standard.asm as well as the DOS batch file asm.bat needed to generate a dump file out of the source files with the Motorola CLAS development kit under DOS. Backup files

of the OS-9 programs are stored here. The source directory contains the sources and header files of the preliminary run control program.

6.4 LINUX-OS-9 SOCKET COMMUNICATION

The following packets are defined (note: all positions are longwords)

Command:	length	opcode	command	value	EOS
Acknowledge:	length	opcode	0	0	EOS
Data:	length	opcode	block#	data	
Error:	length	opcode	errorcode	errmesg	EOS

Valid opcodes are:

opcode	meaning
1	command
2	acknowledgement
3	data
4	error

EOS is defines as 0xefefefef, the errorcode -1 is the only implemented one.
Possible error messages are:

errmesg	meaning
0	unknown command
1	same state
2	invalid value

Possible Commands are:

Command	Flag	Value	Function
0x0	run	0	stop
		1	run
		2	resume
0x1	mode	0	standard
		1	PST learn mode
0x2	save_time	heartbeats	Histogram save interval on the VME CPU
0x4	transfer	x	transfer histo

7 TROUBLE SHOOTING

7.1 VME CRATE

As soon as the VME crate is powered up, the voltage LEDs for +5 V, +12 V and -12 V must be lit, otherwise there is either a problem on the VDIS diagnosis module or the VME crate power supply. In addition the VME clk signal must be driven by the slot 1 module (the Heurikon VME CPU). If the clock LED is not on and the VDIS is fine, the problem is related to the Heurikon.

7.2 DSP PROGRAM DOWNLOAD

The DSP code is downloaded via the VSB master port of the VME CPU. It is transferred via the VSB backplane and the cable connecting the VSB backplane to the DSP front panel. If the veryfy process reports errors you may want to check the cable connection first. In case the setup was moved to another VME crate you also have to make sure, that the VSB backplane is properly installed. If all these things are fine, the problem may be on the VME CPUs VSB master port side or the DSPs VSB slave port side.

8 MAINTENANCE

8.1 CHANGE OF VME ETHERNET ADDRESS

The procedure to change the Heurikons Internet Address is described in **Installing OS-9/ISP Version 1.3 on the V3F and V4F** (which is part of the Heurikon documentation).